

Simple & Compound Interest

COMPLETE LEARNING GUIDE

What You'll Learn

Interest problems test whether a candidate understands that money grows differently depending on whether interest is paid only on the original principal (simple interest) or on the accumulating balance (compound interest) — a distinction that becomes the source of almost every wrong answer on this topic when candidates apply SI logic to a CI question or vice versa. Beyond the basic formulas, placement tests love variations: compounding more frequently than once a year, different interest rates in successive years, and loans repaid through equal installments rather than a single lump sum. This sub-module builds the SI and CI formulas from first principles, covers the fast shortcut formulas for the difference between CI and SI over two and three years (a frequent exam favorite), and works through compounding frequency, variable rates, effective vs nominal rates, and installment-based repayment — each demonstrated with a complete worked calculation so the underlying compounding logic, not just a memorized formula, carries into any disguised version of the question.

1. Simple Interest Basics

Simple interest is calculated only on the original principal for every period, using $SI = (P \times R \times T)/100$, where P is principal, R is the annual rate percent, and T is time in years. The total amount owed or earned is $A = P + SI$. Because the base for interest never changes, SI grows linearly with time — equal amounts of interest accrue in every year.

- Find the simple interest on ₹5000 at 8% per annum for 3 years. $SI = (5000 \times 8 \times 3)/100 = ₹1200$. Amount = $5000 + 1200 = ₹6200$.

2. Compound Interest Basics: Amount and CI Formula

Compound interest is calculated on the principal PLUS all previously accumulated interest, so the base grows every period. Amount $A = P(1 + R/100)^n$, where n is the number of compounding periods, and $CI = A - P$. Because interest earns interest, CI always exceeds SI for the same P, R, and T beyond the first period, and the gap widens every additional year.

- Find the compound interest on ₹5000 at 10% per annum for 2 years, compounded annually. $A = 5000 \times (1.1)^2 = 5000 \times 1.21 = ₹6050$. $CI = 6050 - 5000 = ₹1050$ (compare to SI for the same figures: $5000 \times 10 \times 2 / 100 = ₹1000$ — CI is ₹50 more, purely from interest earning interest in year 2).

3. Difference Between CI and SI: Two-Year and Three-Year Shortcuts

For 2 years, the CI" SI difference equals $P \times (R/100)^2$ — this is exactly the interest earned on the first year's interest during the second year. For 3 years, the shortcut is $CI - SI = P \times R^2 \times (300+R)/10$ v. These formulas let you skip computing both CI and SI in full when a question only asks for the difference.

- $P = ₹8000$, $R = 5\%$, $T = 2$ years. Difference = $8000 \times (5/100)^2 = 8000 \times 0.0025 = ₹20$. Verify: $SI = 8000 \times 5 \times 2/100 = 800$; $CI: A = 8000 \times 1.05^2 = 8820$, $CI = 820$; difference = $820 - 800 = 20$.

- $P = ₹10000$, $R = 10\%$, $T = 3$ years. Difference = $(10000 \times 10^2 \times 310)/10 = (10000 \times 100 \times 310)/1000000 = 310$. Verify: $SI = 10000 \times 10 \times 3/100 = 3000$; $CI: A = 10000 \times 1.1^3 = 13310$, $CI = 3310$; difference = $3310 - 3000 = 310$.

4. Compounding More Than Once a Year

When interest compounds half-yearly, halve the annual rate and double the number of periods; when it compounds quarterly, quarter the rate and quadruple the periods. The formula becomes $A = P(1 + r/100)^n$ • using the ADJUSTED rate and period count, not the original annual figures.

- ₹10000 at 10% per annum, compounded half-yearly, for 1 year: rate per half-year = 5%, periods = 2. $A = 10000 \times (1.05)^2 = 10000 \times 1.1025 = ₹11025$ (higher than annual compounding's ₹11000, since interest compounds more often).

5. Compound Interest With Different Rates in Successive Years

When the rate changes each year, apply each year's growth factor sequentially rather than using a single averaged rate — the amount becomes $P \times (1+r_1/100) \times (1+r_2/100) \times (1+r_3/100) \dots$ for however many years are given.

- ₹1000 invested for 3 years at 5%, then 6%, then 7% in successive years. $A = 1000 \times 1.05 \times 1.06 \times 1.07 = 1000 \times 1.19091 = ₹1190.91$. $CI = ₹190.91$ (using the average rate of 6% instead would incorrectly give $A = 1000 \times 1.06^3 = ₹1191.02$, a close but wrong shortcut).

6. Effective Annual Rate vs Nominal Rate

The nominal (stated) annual rate understates the true annual growth once compounding happens more than once a year. The effective annual rate = $(1 + r/n)^n - 1$, where r is the nominal rate and n is the number of compounding periods per year — this is the rate that would produce the same growth if compounded only once annually.

- A nominal rate of 10% compounded half-yearly has an effective annual rate = $(1 + 0.10/2)^2 - 1 = (1.05)^2 - 1 = 1.1025 - 1 = 0.1025 = 10.25\%$, not 10%.

7. Installments: Repaying a Compound-Interest Loan in Equal Annual Payments

When a loan is repaid through equal annual installments at compound interest, each installment's present value equals $\text{installment} \div (1+r/100)^k$ for the year k it's paid in; the sum of all installments' present values must equal the original loan amount. This requires DISCOUNTING future payments, not simply totaling them.

- A loan is repaid in 2 equal annual installments of ₹6600 each, with interest at 10% per annum. Find the loan amount. $\text{Loan} = 6600/1.1 + 6600/1.1^2 = 6000 + 5454.55 = ₹11454.55$ (simply adding $6600+6600=13200$ and ignoring the time value of money is the classic wrong shortcut here).

Common Pitfalls

- Applying the simple interest formula's flat rate directly to compound interest without accounting for compounding periods.
- Forgetting to halve the rate and double the time period (or quarter/quadruple for quarterly compounding) when compounding happens more than once a year.
- Using the ORIGINAL principal instead of the previous year's amount when rates differ across successive years.
- Confusing nominal rate with effective annual rate, especially in compare-two-schemes questions.
- Applying the 3-year CI" SI difference formula when only 2 years are given (or vice versa) — the two shortcut formulas are not interchangeable.
- In installment problems, simply summing the installment amounts instead of discounting each one back to present value by $(1+r/100)^k$.
- Forgetting that CI always exceeds SI beyond year 1, so a CI answer smaller than the equivalent SI answer signals a calculation error.

Key Takeaways

- $\text{SI} = (P \times R \times T)/100$; $\text{Amount} = P + \text{SI}$.
- CI: $\text{Amount} = P(1+R/100)^n$; $\text{CI} = \text{Amount} - P$.
- CI" SI shortcut for 2 years $= P(R/100)^2$; for 3 years $= P \cdot R^2 \cdot (300+R)/10^5$.
- Half-yearly compounding: halve the rate, double the periods. Quarterly: quarter the rate, quadruple the periods.
- Effective annual rate $= (1+r/n)^n - 1$, always $>$ the nominal rate when $n > 1$.
- For variable yearly rates, multiply successive growth factors — never average the rates.
- Installment repayment: each future installment must be discounted back to present value by dividing by $(1+r/100)$ raised to its year number.

Learn More

GeeksforGeeks — Compound Interest Formula and Problems

<https://www.geeksforgeeks.org/maths/compound-interest/>

IndiaBix — Simple Interest Aptitude Questions

<https://www.indiabix.com/aptitude/simple-interest/>

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